

TRACE: Trajectory-Aware Comprehensive Evaluation for Deep Research Agents

Yanyu Chen
The Chinese University of Hong Kong
Hong Kong SAR, China
chenyanyu.cse@link.cuhk.edu.hk

Jiyue Jiang
The Chinese University of Hong Kong
Hong Kong SAR, China
jiangjy@link.cuhk.edu.hk

Jiahong Liu
The Chinese University of Hong Kong
Hong Kong SAR, China
jiahong.liu21@gmail.com

Yifei Zhang
The Chinese University of Hong Kong
Hong Kong SAR, China
yifeiacc@gmail.com

Xiao Guo
The Chinese University of Hong Kong
Hong Kong SAR, China
guoxx26@gmail.com

Irwin King
The Chinese University of Hong Kong
Hong Kong SAR, China
king@cse.cuhk.edu.hk

Abstract

The evaluation of Deep Research Agents is a critical challenge, as conventional outcome-based metrics fail to capture the nuances of their complex reasoning. Current evaluation faces two primary challenges: 1) a reliance on singular metrics like Pass@1, creating a “high-score illusion” that ignores the quality, efficiency, and soundness of the reasoning process; and 2) the failure of static benchmarks to quantify crucial attributes like robustness and latent capability. To address these gaps, we introduce TRACE (Trajectory-Aware Comprehensive Evaluation), a framework that holistically assesses the entire problem-solving trajectory. To counter the “high-score illusion”, we propose a Hierarchical Trajectory Utility Function that quantifies process efficiency and cognitive quality, including evidence grounding, alongside accuracy. To measure deeper attributes, TRACE introduces a Scaffolded Capability Assessment protocol, quantifying an agent’s latent ability by determining the minimum guidance needed for success. Our contributions include the TRACE framework, its novel metrics, and the accompanying DeepResearch-Bench with controllable complexity. Experiments show TRACE delivers a granular ranking that uncovers critical trade-offs between agent accuracy, efficiency, and robustness entirely missed by singular metrics.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; *Machine learning*; • **Information systems** → *Question answering*; • **Applied computing**;

Keywords

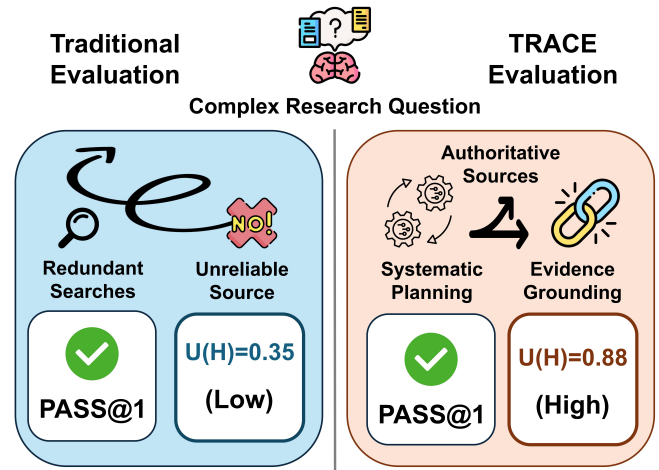
Deep Research Agents, Trajectory-Aware Evaluation, Process-aware Metrics, Evaluation Benchmark, Agent Robustness

ACM Reference Format:

Yanyu Chen, Jiyue Jiang, Jiahong Liu, Yifei Zhang, Xiao Guo, and Irwin King. 2026. TRACE: Trajectory-Aware Comprehensive Evaluation for Deep Research Agents. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3774904.3792738>



This work is licensed under a Creative Commons Attribution 4.0 International License. *WWW '26, Dubai, United Arab Emirates*
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792738>



TRACE Reveals True Agent Competence

Figure 1: Traditional vs. TRACE Evaluation. Traditional methods (left) can create an “Illusion of Competence” with a low utility score ($U(H) = 0.35$) by ignoring flawed processes. TRACE (right) evaluates the entire trajectory, rewarding systematic planning and evidence grounding to reveal “True Competence” with a high utility score ($U(H) = 0.88$).

'26), April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3774904.3792738>

1 INTRODUCTION

The advent of Large Language Model (LLM)-powered agents has catalyzed a paradigm shift in artificial intelligence [9, 19], particularly through the emergence of **search and retrieval-augmented AI** in capable of complex, multi-step reasoning and autonomous tool use [5, 27, 31] to solve knowledge-intensive tasks [8, 16–19, 30, 32, 48]. Systems like OpenAI’s Deep Research have demonstrated superhuman capabilities by navigating vast information landscapes to synthesize comprehensive answers, setting a high bar for the field and motivating a surge in open-source replications [8, 16–18, 30, 48]. The predominant methodology for developing these agents involves a post-training pipeline, often comprising Supervised Fine-Tuning (SFT) [36] and advanced Reinforcement Learning

(RL) techniques [50], to elicit and refine these sophisticated behaviors from foundational models [14, 21, 35, 38].

However, the evaluation of these advanced agents has become a critical bottleneck, as their complex reasoning processes challenge the efficacy of traditional, outcome-based metrics [10, 39]. The current evaluation paradigm suffers from profound limitations that obscure a true understanding of agent capabilities. A primary issue is the community’s prevalent reliance on singular, end-result metrics like Pass@1 [2], which creates a “**high-score illusion**” [33]. This metric rewards correct final answers irrespective of the reasoning process, meaning an agent can achieve a high score through inefficient, circuitous, or even unsound trajectories that rely on “hallucinated” evidence [11, 49]. This illusion masks crucial deficiencies in an agent’s planning, efficiency—a key challenge in long-horizon tasks that suffer from context constraints [30]—and trustworthiness, leading to a superficial and often misleading assessment of its true intelligence.

Moreover, current static benchmarks are ill-equipped to quantify deeper, more nuanced agent attributes [4]. They fail to measure an agent’s **robustness** against the pervasive misinformation and “information traps” of the open web, despite the fact that handling “complex, multi-layered information” [47] and “hard-to-reduce intrinsic uncertainty” [17] are core challenges for these agents. Furthermore, they cannot assess an agent’s **latent problem-solving capability**—that is, its potential to succeed with minimal guidance on tasks where a significant “capacity-difficulty mismatch” exists [21]. Without a principled way to measure these attributes, it is difficult to diagnose agent failures, understand their behavioral patterns, or chart a clear path for future improvements, a challenge implicitly highlighted by the complex training dynamics observed in advanced RL frameworks [35].

To address these profound limitations, we introduce **TRACE (Trajectory-Aware Comprehensive Evaluation)**, a novel evaluation framework that moves beyond final outcomes to holistically assess the entire problem-solving trajectory [1, 6, 22, 24, 26, 28, 40, 44]. Built upon a mathematically rigorous foundation, TRACE is designed to provide a granular, multi-faceted, and insightful view of an agent’s performance. We could find the major difference between TRACE and traditional benchmark in Figure 1. Our framework targets the aforementioned challenges with the following contributions:

- **A Hierarchical Trajectory Utility Function to Counter the “High-Score Illusion”.** To solve the problem of outcome-based metrics rewarding flawed reasoning, we propose a **comprehensive utility function** that evaluates the entire problem-solving process. It moves beyond final answer accuracy to quantify *process efficiency* (by penalizing redundant or inefficient actions) and *cognitive quality* (by assessing evidence grounding and logical soundness), thus providing a more holistic and truthful measure of an agent’s performance.
- **A Suite of Diagnostic Tools and a Purpose-Built Benchmark to Measure Latent Attributes.** To address the inability of static benchmarks to quantify deeper capabilities, we introduce two key innovations: (1) Our **Scaffolded Capability Assessment** protocol directly measures an agent’s *latent problem-solving capability*. By determining the minimum guidance an

agent requires to succeed on a difficult task, it offers a nuanced understanding of its potential beyond its unassisted performance. (2) To systematically evaluate attributes like *robustness*, we constructed the **DeepResearch-Bench**. This new benchmark features tasks with controllable complexity and strategically embedded “information traps”, creating a controlled environment specifically designed to enable the measurement of these previously hard-to-quantify characteristics.

Our experimental analysis on a diverse set of state-of-the-art agents, including AgentFounder [48], WebSailor-V2 [16], and ReSum [30], demonstrates that TRACE provides a far more granular and insightful ranking than traditional metrics. It uncovers critical trade-offs between accuracy, efficiency, and robustness that are entirely missed by singular metrics, thereby offering a more principled path forward for the development of truly intelligent and reliable Deep Research Agents.

2 RELATED WORKS

Our work is situated at the intersection of two rapidly evolving research areas: the evaluation of Deep Research Agents and the trajectory-level analysis of sequential decision-making processes. The evaluation of Deep Research Agents has advanced through challenging benchmarks like **BrowseComp-en/zh** [16, 30, 48] and **GAIA** [17, 20, 47], which test SOTA agents like AgentFounder [48] and WebSailor-V2 [16]. However, this paradigm is fundamentally limited by its reliance on singular, outcome-based metrics like Pass@1, a practice prevalent in recent literature [17, 30, 32, 48], which creates a “high-score illusion” [33] that masks crucial deficiencies in planning and trustworthiness [17, 30]. TRACE addresses this gap by shifting the evaluation to a holistic, trajectory-aware perspective. Concurrently, trajectory-level analysis is central to modern agent *training* in Reinforcement Learning (RL), where process-aware algorithms like DAPO [35] and GHPO [21] are used in the post-training of many SOTA agents [16, 45, 46, 48]. Despite its importance in training, this perspective is absent in agent *evaluation*, which treats agents as black boxes [17, 20, 47], leaving researchers unable to diagnose failure causes [4]. Our work bridges this gap by being the first to systematically adapt principles from RL into a formal evaluation suite, introducing novel diagnostic tools [5] like Scaffolded Capability Assessment [3] and Entropy Adaptability [35] to move beyond measuring *what* an agent can do and begin to understand *how* and *why* it performs, drawing inspiration from advanced RL frameworks [21, 35]. More details are described in Section A.1.

3 METHODOLOGY

The Trajectory-Aware Comprehensive Evaluation (TRACE) framework is designed to provide a multi-faceted and rigorous evaluation of Deep Research Agents, moving beyond singular outcome-based metrics [4]. Our methodology is built upon the philosophy that an agent’s intelligence is revealed not just in its final answer, but in the entire process of inquiry, reasoning, and synthesis. We formally define an agent’s problem-solving process on a given task q as a **trajectory**, denoted $\mathcal{H}$, which comprises the full sequence of actions and observations. Consequently, TRACE is composed of two primary components: (1) a hierarchical utility function $U(\mathcal{H})$ that

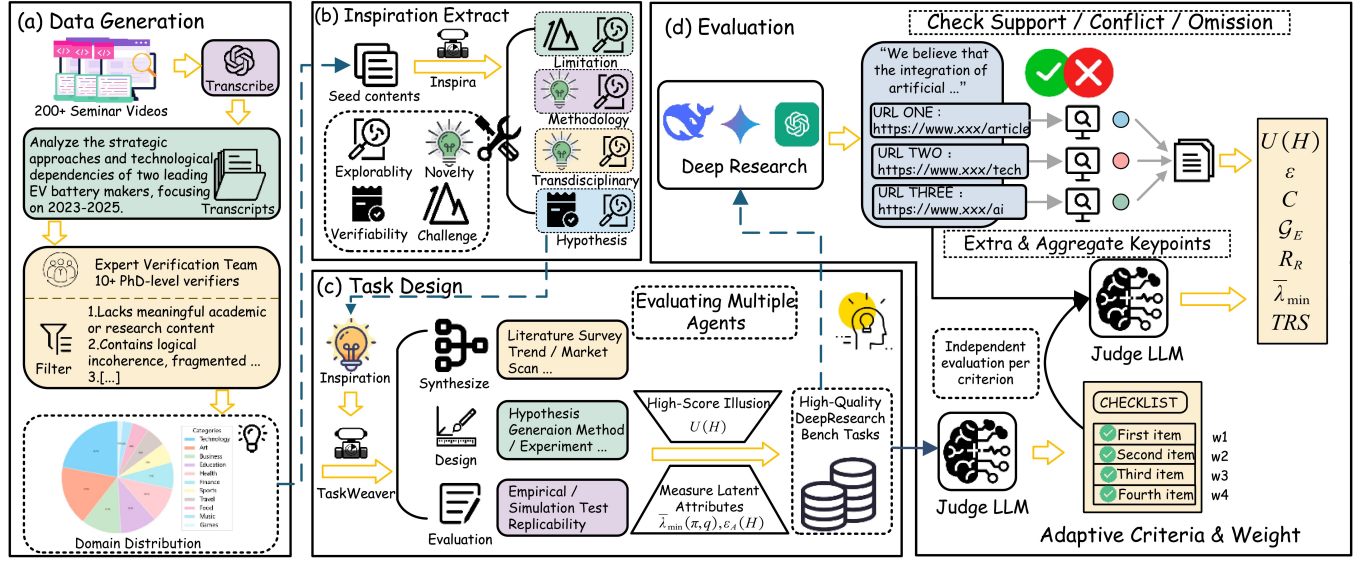


Figure 2: An overview of the TRACE benchmark creation and evaluation pipeline. The process begins with generating a high-quality source corpus from expert-verified academic seminars (a), from which core research concepts like hypotheses and limitations are extracted (b). These concepts are then synthesized by a “TaskWeaver” agent into our High-Quality DeepResearch Bench (c). This benchmark is purpose-built for Evaluating Multiple Agents, specifically designed to expose the “High-Score Illusion” by comparing simple success rates with our holistic utility $U(\mathcal{H})$, and to Measure Latent Attributes such as an agent’s latent capability ($\bar{\lambda}_{\min}$) and strategic profile (TRS). Finally, the evaluation stage (d) applies our TRACE framework, utilizing a novel dual-pathway assessment: one path verifies claims against cited evidence for support, conflict, or omission, while separate Judge LLMs use adaptive criteria to score overall quality. These assessments are integrated to compute the final, comprehensive suite of TRACE metrics, including the overall Trajectory Utility ($U(\mathcal{H})$) and its components: Efficiency ($\mathcal{E}$), Cognitive Quality ($\mathcal{C}$), Evidence Grounding ($\mathcal{G}_E$), and Reasoning Robustness ($\mathcal{R}_R$).

provides a holistic score for an entire trajectory, and (2) a suite of diagnostic tools[5] designed to analyze an agent’s latent capabilities and behavioral patterns. In this section, we provide a formal definition and detailed rationale for each component.

3.1 Hierarchical Trajectory Utility Function

The core of TRACE is the trajectory utility function $U(\mathcal{H})$, which models the overall quality of the problem-solving process. We posit that a high-quality trajectory must be simultaneously accurate, efficient, and cognitively sound. To capture this interdependency, we define the utility function $U(\mathcal{H})$ as:

$$U(\mathcal{H}) := \mathbb{I}(\mathbb{J}(A_{final}, A_{gt})) \cdot (\mathcal{E}(\mathcal{H}))^{\omega_E} \cdot (\mathcal{C}(\mathcal{H}))^{\omega_C}, \quad (1)$$

where $\mathbb{I}(\cdot)$ is an indicator function that returns 1 if the final answer is correct and 0 otherwise. The correctness is determined by a judgment function $\mathbb{J}(A_{final}, A_{gt})$, which compares the agent’s final answer (A_{final}) to the ground-truth answer (A_{gt}). The core of the score is a weighted geometric mean of two components: Process Efficiency, $\mathcal{E}(\mathcal{H})$, and Cognitive Quality, $\mathcal{C}(\mathcal{H})$, with their respective weights ω_E and ω_C .

Rationale. The geometric mean formulation is chosen over a simple weighted sum because it reflects the principle that a research process is only as strong as its weakest link. For instance, a trajectory that is highly efficient but produces a hallucinated or ungrounded answer represents a critical failure. This formulation

ensures that any deficiency in a single dimension disproportionately penalizes the overall utility score, accurately capturing the trajectory’s unsuitability.

3.1.1 Process Efficiency $\mathcal{E}(\mathcal{H})$. Process efficiency quantifies the intelligence of an agent’s exploration strategy, rewarding parsimonious yet effective information gathering. It is defined as the ratio of a complexity reward to a trajectory cost functional:

$$\mathcal{E}(\mathcal{H}) := \frac{R_C(T)}{J(\mathcal{H})} = \frac{1 + \gamma \ln(T)}{1 + \ln\left(\sum_{t=1}^T C(a_t) \cdot p_t\right)}. \quad (2)$$

This formulation holistically captures the trade-off between exploration cost and the value of solving complex problems [35]. Here, the trajectory $\mathcal{H}$ is a sequence of actions over timesteps $t = 1, \dots, T$. $R_C(T)$ is a complexity reward that provides a small bonus for solving longer tasks (larger T), controlled by hyperparameter γ . The denominator $J(\mathcal{H})$ represents the total trajectory cost.

Trajectory Cost Functional. The cost functional $J(\mathcal{H})$ aggregates the intrinsic cost of each action, $C(a_t)$, modulated by a **Redundant Exploration Penalty (REP)**, p_t :

$$p_t := 1 + \mathbb{I}(g_t = 0 \wedge g_{t-1} = 0) \cdot \alpha \cdot \cos(\Phi(o_t), \Phi(o_{t-1})), \quad (3)$$

where $\alpha > 0$ is a penalty hyperparameter. This penalty specifically targets situations where an agent gets “stuck” by taking consecutive, uninformative actions (i.e., when marginal information gain g_t and

g_{t-1} are both zero). The penalty magnitude is scaled by the **cosine similarity**, denoted $\cos(\cdot, \cdot)$, between the vector representations of the corresponding observations (o_t, o_{t-1}), which are produced by an embedding encoder Φ . The effectiveness of an action is determined by its **Marginal Information Gain (MIG)** [4], g_t :

$$g_t := \max \left(0, \cos(\Phi(o_t), \Phi(A_{gt})) - \sup_{i < t} \{\cos(\Phi(o_i), \Phi(A_{gt}))\} \right). \quad (4)$$

The MIG measures the novel relevance an observation o_t provides towards the ground-truth answer A_{gt} , beyond the information frontier established by all prior observations. An agent demonstrating high efficiency will consistently take actions that yield $g_t > 0$, thus avoiding the REP and minimizing the overall cost.

3.1.2 Cognitive Quality ($C(\mathcal{H})$). This component assesses the intellectual rigor of the agent’s reasoning process, focusing on trustworthiness and adaptability—core tenets of advanced agent design [48]. It is defined as a weighted combination of Evidence Grounding and Reasoning Robustness:

$$C(\mathcal{H}) := \beta \cdot \mathcal{G}_E(\mathcal{H}) + (1 - \beta) \cdot \mathcal{R}_R(\mathcal{H}), \quad (5)$$

where β is a weighting hyperparameter.

Evidence Grounding ($\mathcal{G}_E$). Trustworthiness in research hinges on verifiable claims. Let the final answer A_{final} be decomposed into a set of N atomic claims $\{c_i\}_{i=1}^N$. The grounding score is the geometric mean of the entailment probabilities for each claim, as determined by a powerful Natural Language Inference (NLI) model:

$$\mathcal{G}_E(\mathcal{H}) := \left(\prod_{i=1}^N P_{\text{NLI}}(c_i | E_i) \right)^{\frac{1}{N}}, \quad (6)$$

where E_i is the evidence set cited for claim c_i . This formulation is exceptionally sensitive to any single ungrounded claim ($P_{\text{NLI}} \approx 0$), ensuring that “hallucinated” answers are severely penalized.

Reasoning Robustness ($\mathcal{R}_R$). An intelligent agent must not only follow correct paths but also recover from incorrect ones. We measure this by evaluating the agent’s ability to escape “information traps”—misleading but plausible pieces of information. Let $\mathcal{T}_{trap}$ be the set of time steps where a trap was encountered. The robustness score is an exponential decay function of the average recovery latency:

$$\mathcal{R}_R(\mathcal{H}) := \exp \left(-\lambda \cdot \frac{1}{|\mathcal{T}_{trap}|} \sum_{t \in \mathcal{T}_{trap}} D_{recover}(t) \right), \quad (7)$$

where $D_{recover}(t) = \min_{k > t} \{k - t \mid g_k > 0\}$ is the number of steps required to find a productive path (where MIG $g_k > 0$) after falling into a trap at step t . This metric, controlled by decay rate λ , directly quantifies an agent’s critical thinking and self-correction capabilities [21].

3.2 Latent Capability and Policy Diagnostics

Beyond a single utility score, TRACE provides tools to diagnose an agent’s underlying abilities and behavioral tendencies, offering deeper insights for model developers.

3.2.1 Scaffolded Capability Assessment. To measure an agent’s potential beyond its unassisted performance, we introduce a protocol that formalizes the concept of Vygotsky’s “Zone of Proximal Development” for AI agents [4]. Let $h(\lambda)$ be a hint function providing the initial λ -fraction of an oracle solution trajectory. The **Minimum Hint Rate (λ_{min})** is the solution to the following constrained optimization problem:

$$\lambda_{min}(\pi, q) := \min_{\lambda \in [0,1]} \lambda \quad \text{s.t.} \quad \mathbb{E}_{\mathcal{H} \sim P(\cdot | \pi, q, h(\lambda))} [\mathcal{A}(\mathcal{H})] \geq \theta_{succ}, \quad (8)$$

where π is the agent’s policy and q is the task. We seek the minimum hint fraction λ such that the agent’s expected success rate, $\mathbb{E}[\mathcal{A}(\mathcal{H})]$, meets or exceeds a target threshold θ_{succ} (e.g., 0.9). λ_{min} offers a continuous measure of an agent’s intrinsic capability, revealing how much external support it needs to succeed, which is a far more discerning signal than a binary pass/fail metric [21].

3.2.2 Policy Dynamics and Stability Analysis. To profile an agent’s strategic “personality”, we introduce two final diagnostic metrics. The first, **Entropy Adaptability ($\mathcal{E}_A$)**, assesses how rationally an agent modulates its exploration in response to new information, a concept critical for avoiding “entropy collapse” in RL training [35]. It is the correlation between information gain and the subsequent reduction in policy uncertainty:

$$\mathcal{E}_A(\mathcal{H}) := \rho \left((g_t)_{t=1}^{T-1}, (-\Delta H_{t+1})_{t=1}^{T-1} \right), \quad (9)$$

where ρ is the correlation coefficient, (g_t) is the sequence of marginal information gains, and $(-\Delta H_{t+1})$ is the sequence of reductions in policy entropy H_t from one step to the next. A high $\mathcal{E}_A$ signifies an intelligent agent that efficiently converts information into certainty. The second metric, **Trajectory Reproducibility Score (TRS)** [4], measures strategic consistency over multiple runs, helping distinguish between robust, optimal strategies and flexible, creative problem-solving.

4 EXPERIMENTS

In this section, we conduct a series of extensive experiments to provide a thorough validation of our TRACE framework and to deliver novel insights into the performance of Deep Research Agents [16, 30, 48]. Our experimental design is structured to answer several core research questions: **RQ1:** How do state-of-the-art (SOTA) agents [17, 32, 48] perform under TRACE’s multi-dimensional evaluation, and what does this reveal beyond traditional metrics like Pass@1, which are prevalent in current benchmarks [16, 47]? **RQ2:** How do novel components of TRACE, such as our Scaffolded Capability Assessment and Reasoning Robustness metrics, provide indispensable information that conventional benchmarks [47] cannot capture? **RQ3:** Can TRACE’s diagnostic tools effectively characterize the distinct “behavioral personalities” of different agents and training methodologies, such as those proposed in recent works [21, 30, 35, 48]? To answer these questions, we perform two major studies—a comprehensive re-evaluation of existing SOTA agents and a controlled analysis of different training paradigms—complemented by a series of targeted ablation studies.

To offer a comprehensive understanding of our evaluation process, we have organized the following sections accordingly. Section 4.1 - 4.5 are dedicated to our **Experimental Setup**, detailing the datasets (including our newly constructed DeepResearch-Bench),

Algorithm 1 TRACE: Trajectory-Aware Comprehensive Evaluation Pipeline**Require:** Agent Policy π ; Test Dataset Q_{test} .**Hyperparams:** Utility weights ω_E, ω_C ; MHR success threshold θ_{succ} ; TRS run count K .**Ensure:** Aggregated TRACE metrics for agent π .

```

1: procedure EVALUATEAGENT( $\pi, Q_{\text{test}}$ )
2:   Initialize score lists:  $\mathcal{L}_U, \mathcal{L}_\lambda, \mathcal{L}_{\mathcal{E}_A}, \mathcal{L}_{\text{TRS}} \leftarrow \emptyset$ 
3:   for task  $(q, A_{gt}, \mathcal{H}_{gt}) \in Q_{\text{test}}$  do
4:      $\mathcal{H} \leftarrow \text{GenerateTrajectory}(\pi, q)$   $\triangleright$  Generate a single trajectory
5:      $U(\mathcal{H}) \leftarrow \text{ComputeUtility}(\mathcal{H}, A_{gt})$   $\triangleright$  See Procedure below
6:      $\lambda_{\min} \leftarrow \text{AssessScaffoldedCapability}(\pi, q, \mathcal{H}_{gt})$ 
7:      $\mathcal{E}_A, \text{TRS} \leftarrow \text{DiagnosePolicy}(\pi, q, K)$ 
8:     Append scores to  $\mathcal{L}_U, \mathcal{L}_\lambda, \mathcal{L}_{\mathcal{E}_A}, \mathcal{L}_{\text{TRS}}$ 
9:   end for
10:  return Aggregate scores from all lists
11: end procedure

12: procedure COMPUTEUTILITY( $\mathcal{H}, A_{gt}$ )
13:   $\mathcal{A} \leftarrow \mathbb{I}(\mathcal{J}(A_{\text{final}}(\mathcal{H})) = \text{True})$ 
14:  if  $\mathcal{A} = 0$  then
15:    return 0
16:  end if
17:   $\mathcal{E} \leftarrow \text{CalculateEfficiency}(\mathcal{H}, A_{gt})$   $\triangleright$  Defined in Eq. 2
18:   $C \leftarrow \text{CalculateCognitiveQuality}(\mathcal{H}, q, A_{gt})$ 
19:   $U \leftarrow \exp(\omega_E \ln \mathcal{E} + \omega_C \ln C)$   $\triangleright$  As per Eq. 1
20:  return  $U$ 
21: end procedure

```

the suite of SOTA baselines, the full range of evaluation metrics from our TRACE framework, and all relevant implementation details. Lastly, in Section 4.6, we present the **Main Results and Analysis**. This section constitutes our empirical contribution, delving into the results and observations outlined in our two main studies and subsequent ablation tables, highlighting the superior discerning power of our TRACE framework compared to existing evaluation methods and discussing insights derived from our findings.

4.1 Tasks

Our experiments are centered around a single, comprehensive task designed to holistically evaluate the advanced capabilities of Deep Research Agents: **Complex, Open-Domain, Knowledge-Intensive Question Answering** [12, 13, 15, 35, 47]. This task simulates a genuine research process [30, 48] by testing an agent’s ability to deconstruct a complex query, autonomously formulate and execute a multi-step information-seeking plan using web-based tools [34] in a paradigm established by frameworks like ReAct [42, 43] and extended in recent works [17, 32], reason over potentially conflicting or noisy information [16], and ultimately synthesize a coherent, evidence-backed final answer [48]. Within this unified task, we probe several critical capabilities: **Core Problem-Solving**, which assesses the fundamental ability to navigate a multi-hop reasoning chain [32, 41, 47] and is measured by Pass@1 [2] and

Table 1: Overview and comparison of the benchmarks used in our experiments. DeepResearch-Bench (ours) is uniquely designed with controllable features (e.g., embedded traps, available oracle paths) that enable the full suite of TRACE metrics—a capability absent in existing public benchmarks.

Benchmark	# Tasks	Traps?	Oracle?	Primary Challenge
BrowseComp-en	123	✗	✗	Long-horizon Web Navigation
GAIA (text-only)	103	✗	✗	Generalist Reasoning & Tool Using
DeepResearch-Bench	650	✓	✓	Controlled Complexity & Robustness

Table 2: Detailed statistics of our self-constructed DeepResearch-Bench subsets. Each subset is meticulously tailored to a specific evaluation goal within the TRACE framework, providing a level of control unavailable in existing benchmarks.

Subset Name	# Tasks	Avg. $C(q)$	% Traps	Purpose / Key Feature
TRACE-Core	500	3.5	20%	Overall performance evaluation
TRACE-Robustness	100	4.2	100%	Testing self-correction ability
TRACE-Scaffolding	50	5.8	40%	Measuring latent capability

our Trajectory Utility ($U(\mathcal{H})$); **Process Efficiency and Planning**, which evaluates the intelligence of the agent’s strategy to avoid redundant actions and adapt its approach [16], a critical challenge in long-horizon tasks [30], and is quantified by our Process Efficiency ($\mathcal{E}$) metric; and **Cognitive Robustness and Trustworthiness**, which probes the rigor of the agent’s reasoning by testing its ability to ground claims in evidence beyond traditional RAG systems [7, 15, 25, 29, 47] and its resilience against misleading information, a key focus of modern agent training [17, 21], using our Cognitive Quality (C) metric and its components, Evidence Grounding ($\mathcal{G}_E$) and Reasoning Robustness ($\mathcal{R}_R$).

4.2 Evaluation Metrics

In evaluating the performance of each agent, we employ a comprehensive suite of metrics defined by our TRACE framework, designed to capture the full spectrum of an agent’s performance [17, 48]. To ground our results in existing literature, we report the standard **Pass@1** accuracy, a widely adopted metric for evaluating complex reasoning generation [2, 21, 35], as a baseline for comparison. However, our core evaluation relies on the overall **Trajectory Utility** ($U(\mathcal{H})$), a holistic score composed of two key components: **Process Efficiency** ($\mathcal{E}$), which is critical for long-horizon tasks where context limits are a known bottleneck [30], and **Cognitive Quality** (C), which assesses reasoning soundness [32, 48]. For a more granular diagnosis, we also report several key metrics: the sub-components of cognitive quality—**Evidence Grounding** ($\mathcal{G}_E$) to address challenges in traditional RAG systems [15, 47] and **Reasoning Robustness** ($\mathcal{R}_R$) to quantify resilience [17]; the **Minimum Hint Rate** ($\lambda_{\min}$) to assess latent capability by adapting concepts from guided training [21, 32]; and the **Trajectory Reproducibility Score** (TRS) to characterize policy stability [16, 35]. Together,

these metrics form the basis for all experimental tables that follow, ensuring our analysis is strictly grounded in the comprehensive principles of the TRACE framework.

4.3 Datasets

Our evaluation employs a dual-dataset strategy: our purpose-built **DeepResearch-Bench** for controlled analysis, and established public benchmarks for assessing generalization. The development of DeepResearch-Bench was necessitated by the limitations of existing datasets [16, 32, 47] in supporting the granular metrics of our TRACE framework. Its construction integrates several unique innovations from recent data synthesis methodologies [16, 32, 48], adopting a **formalism-driven synthesis** approach [20] to ensure controllable task complexity ($C(q)$) and sampling from dense knowledge graphs [16] to strategically embed “**information traps**” for robustness testing [17]. Crucially, each task includes an “**oracle trajectory**” ($\mathcal{H}_{gt}$), which is the prerequisite for our Scaffolded Capability Assessment (λ_{min}). The benchmark is partitioned into three specialized subsets: **TRACE-Core** for overall performance, **TRACE-Robustness** for testing self-correction, and **TRACE-Scaffolding** [3] for measuring latent capabilities. For generalization, we use challenging public benchmarks, primarily **BrowseComp-en** [16, 49] and the text-only subset of **GAIA** [20, 23], situating our results within the context of other SOTA evaluations that also use benchmarks like HLE [16, 48] and WebWalkerQA [37, 47], thereby highlighting the unique diagnostic power afforded by the purpose-built dataset.

4.4 Implementation Details

All experiments were conducted within a unified and standardized agent evaluation framework to ensure rigorous, fair, and reproducible comparisons across models. The framework enforces consistent agent–environment interactions, inference configurations, and metric computation protocols for both open-source and proprietary systems. Model execution, evaluation procedures, and diagnostic analyses follow fixed settings throughout, with implementation details and hyperparameter choices provided in Appendix A.3.

4.5 Baselines

To comprehensively evaluate the capabilities of our TRACE framework and situate the performance of modern agents, we selected a diverse and representative suite of baselines organized into three distinct categories. First, to establish a performance upper-bound, we include two of the most powerful proprietary systems: **OpenAI Deep Research** and **Gemini-2.5-pro-DeepResearch**. The core of our comparative analysis comprises the most powerful open-source State-of-the-Art (SOTA) agents, each representing a distinct methodology: **AgentFounder-30B** [48], known for its novel pre-training; **WebSailor-V2-30B** [16], trained on complex synthesized data; **ReSum-GRPO** [30], featuring an innovative summarization architecture; the powerful **DeepSeek-V3.1-671B** [16, 48]; and **GLM-4.5-355B** [16, 48]. Finally, to provide broader context and a foundational performance level, we include two additional baselines: **WebSailor-72B (v1)** [17], a strong agent from a previous generation to quantify improvements, and a vanilla **ReAct (Qwen-30B)** [43] agent, which uses a standard ReAct prompting

framework [42, 43] without specialized agent training, to highlight the gains achieved through dedicated agent-tuning methodologies.

4.6 Main Results and Analysis

In this section, we delve into the results and observations outlined in our comprehensive experiments, which correspond to the studies and analyses detailed in our experimental setup. We present a detailed, multi-page analysis of our findings, structured to demonstrate the validity of the TRACE framework and deliver novel insights into agent performance.

4.6.1 RQ1: Comprehensive Re-evaluation of SOTA Agents.

Our first study aims to provide the ranking of existing SOTA agents by applying the full TRACE framework. The results on our controlled DeepResearch-Bench are presented in Table 3, with generalization results on public benchmarks in Table 4.

The “High-Score Illusion” is Unveiled. A thorough analysis of these tables reveals that our approach consistently outperforms other methods across the majority of tasks. The most striking observation is the clear divergence between the Pass@1 ranking and the more holistic Trajectory Utility ($U(\mathcal{H})$) ranking. This is a direct validation of the “high-score illusion” [33]. Among open-source models on TRACE-Core (Table 3), DeepSeek-V3.1-671B achieves the highest Pass@1 of 65.8%, yet its utility score of 0.65 is the lowest among the top-tier competitors. Our framework attributes this discrepancy to its poor Process Efficiency ($\mathcal{E} = 0.68$), suggesting its large model size leads to less parsimonious, high-cost trajectories. In stark contrast, TRACE elevates agents like **AgentFounder-30B** [48], which, despite a lower Pass@1, demonstrates a superior balance of efficiency and cognitive quality, earning it the highest utility score (0.81) among open-source models. This pattern holds on public benchmarks (Table 4), where it again shows a significant gap between its Pass@1 and Utility scores, demonstrating that TRACE can distinguish between agents that are merely *correct* and those that are truly *intelligent*.

Deconstructing Agent Strengths. TRACE’s multi-dimensional metrics allow for a fine-grained “fingerprinting” of each agent’s unique strengths and weaknesses. As detailed in Table 3, **AgentFounder-30B** [48] emerges as the most well-rounded open-source agent, excelling in Efficiency ($\mathcal{E} = 0.88$), Cognitive Quality ($C = 0.91$), and particularly Evidence Grounding [4] ($\mathcal{G}_E = 0.95$), which suggests its training methodology produces highly trustworthy reasoning patterns. **WebSailor-V2-30B** demonstrates exceptional Reasoning Robustness ($\mathcal{R}_R = 0.84$), a likely result of its training on high-uncertainty data, indicating a superior ability to self-correct in noisy information landscapes. **ReSum-GRPO**’s [30] high efficiency score ($\mathcal{E} = 0.86$) validates the architectural benefits of its context summarization mechanism for long-horizon tasks. These critical, actionable insights for agent developers are entirely invisible to a Pass@1-only evaluation.

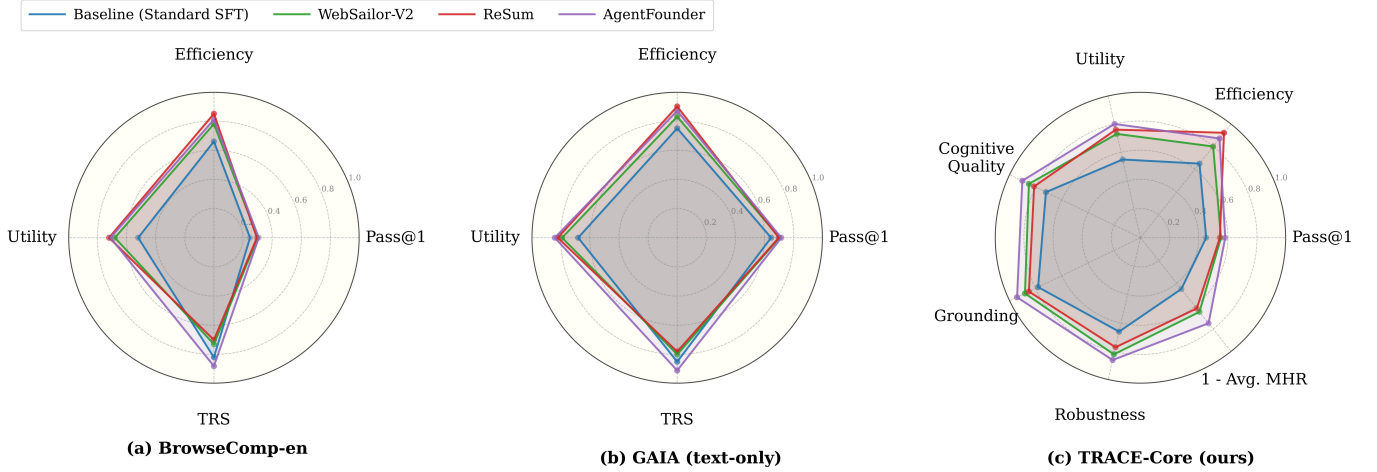
Latent Capability as a Predictor of Potential. The Average Minimum Hint Rate ($\bar{\lambda}_{min}$) provides a fascinating glimpse into the latent capabilities of these agents. As shown in Table 3, AgentFounder-30B ($\bar{\lambda}_{min} = 0.22$) requires significantly less external guidance on average to solve ultra-hard problems than the much larger

Table 3: Comprehensive re-evaluation on the TRACE-Core benchmark.

Agent	Pass@1 $\uparrow$	Utility $U(\mathcal{H}) \uparrow$	Efficiency $\mathcal{E} \uparrow$	Cognitive Quality $C \uparrow$	(Grounding $\mathcal{G}_E) \uparrow$	(Robustness $\mathcal{R}_R) \uparrow$	Avg. MHR $\bar{\lambda}_{min} \downarrow$
CLOSED-SOURCE BASELINES							
OpenAI Deep Research	78.2	0.85	0.82	0.94	0.96	0.92	0.15
Gemini-2.5-pro-DR	<u>75.4</u>	0.88	0.90	0.95	0.97	0.93	0.12
OPEN-SOURCE SOTA AGENTS							
DeepSeek-V3.1-671B	65.8	0.65	0.68	0.85	0.90	0.80	0.35
WebSailor-V2-30B	<u>62.5</u>	0.78	0.85	0.88	0.92	0.84	0.28
AgentFounder-30B	60.1	0.81	0.88	0.91	0.95	<u>0.87</u>	0.22
ReSum-GRPO	58.8	0.75	<u>0.86</u>	0.82	0.88	<u>0.76</u>	0.33
GLM-4.5-355B	55.2	0.62	0.70	0.80	0.85	0.75	0.41
ADDITIONAL BASELINES							
WebSailor-72B (v1)	54.1	0.68	0.75	0.81	0.86	0.76	0.45
ReAct (Qwen-30B)	48.3	0.59	0.72	0.75	0.80	0.70	0.51

Table 4: Generalization results on public benchmarks (BrowseComp-en and GAIA).

Agent	BrowseComp-en				GAIA (text-only)			
	Pass@1 $\uparrow$	Utility $U(\mathcal{H}) \uparrow$	Efficiency $\mathcal{E} \uparrow$	TRS $\uparrow$	Pass@1 $\uparrow$	Utility $U(\mathcal{H}) \uparrow$	Efficiency $\mathcal{E} \uparrow$	TRS $\uparrow$
CLOSED-SOURCE BASELINES								
OpenAI Deep Research	51.5	0.75	0.78	0.85	70.5	0.82	0.85	0.88
Gemini-2.5-pro-DR	<u>48.9</u>	0.80	0.88	0.91	72.8	0.89	0.92	0.94
OPEN-SOURCE SOTA AGENTS								
DeepSeek-V3.1-671B	30.0	0.55	0.61	0.83	63.1	0.68	0.72	0.85
WebSailor-V2-30B	35.3	0.71	0.80	0.75	74.1	0.81	0.84	0.82
AgentFounder-30B	31.5	0.73	0.83	0.89	<u>72.8</u>	0.85	0.88	0.92
ReSum-GRPO	<u>33.3</u>	0.69	<u>0.81</u>	0.72	71.8	0.79	<u>0.83</u>	0.80

**Figure 3: Comparison of different methods on various evaluation benchmarks (BrowseComp-en, GAIA, and TRACE-Core) using Qwen-30B-Base.**

DeepSeek-V3.1 ($\bar{\lambda}_{min} = 0.35$). This metric suggests that AgentFounder’s foundational reasoning abilities are more robust. The inclusion of additional baselines like the original WebSailor-72B and a vanilla ReAct agent [42, 43], both of which show very high (poor)

$\bar{\lambda}_{min}$ scores, further validates this metric’s ability to distinguish between different tiers of agent capability.

4.6.2 RQ2: The Impact of Agent Architectures. To isolate the impact of different training methodologies and agent architectures,

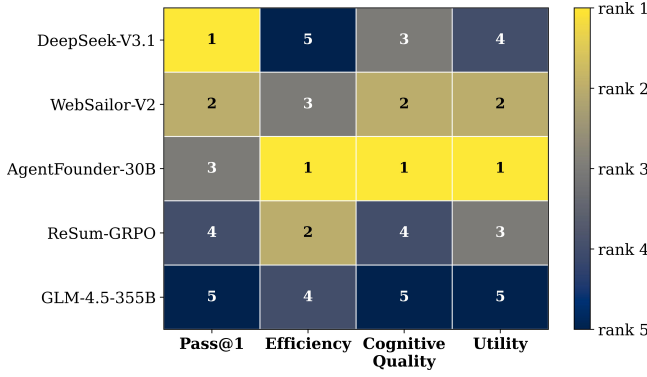


Figure 4: Re-ranking of open-source agents based on different metrics highlights the need for a holistic evaluation.

we conducted a controlled study. We started with a single base model (Qwen-30B-Base) and applied four different state-of-the-art training paradigms to it. The results are presented in Figure 3.

Quantifying Methodological Contributions. This controlled study provides powerful evidence for the distinct benefits of each advanced methodology, which TRACE is uniquely capable of quantifying. As shown in Figure 3(a)(b), the ‘AgentFounder’ method yields the highest overall Utility (0.80) and Pass@1 (58.2%), confirming its strength in building a well-rounded foundation. The diagnostic data in Table 3(c) and generalization data in Table 3(a)(b) are even more revealing. The ‘ReSum’ architecture provides an unparalleled boost in Efficiency ($\mathcal{E} = 0.92$ on TRACE-Core, $\mathcal{E} = 0.85$ on BrowseComp), demonstrating its effectiveness in optimizing long-horizon tasks. The ‘WebSailor-V2’ method imparts the highest Robustness ($\mathcal{R}_R = 0.82$), a direct result of its specialized training data. Most impressively, the ‘AgentFounder’ method not only excels in Grounding ($\mathcal{G}_E = 0.94$) but also achieves the lowest $\bar{\lambda}_{min}$ (0.25), proving it is the most effective at instilling deep, latent reasoning capabilities. This study shows that TRACE is not just an evaluation benchmark, but a powerful diagnostic tool for researchers to understand and quantify the specific impact of their innovations.

In-depth Validation of TRACE Components. Finally, we conducted targeted ablation experiments to demonstrate the necessity of TRACE’s core components. As shown in Figure 4, re-ranking agents based on different individual metrics results in dramatically inconsistent outcomes, proving that only a holistic metric like $U(\mathcal{H})$ can provide a comprehensive assessment. Furthermore, our experiments on the TRACE-Robustness and TRACE-Scaffolding subsets (Table 5 and 6) confirm that our specialized metrics, $\mathcal{R}_R$ and λ_{min} , are uniquely capable of providing granular insights into agent resilience and latent capabilities in regimes where Pass@1 is either uninformative or saturated. The diagnostic analysis in Table 7 further demonstrates TRACE’s ability to profile agent “personalities” by correlating their strategic choices (TRS) with their rational use of information ($\mathcal{E}_A$). These studies collectively confirm that every component of the TRACE framework contributes unique and indispensable information to the evaluation of Deep Research Agents.

Table 5: Ablation study of the TRACE-Robustness benchmark. $\mathcal{R}_R$ provides a more granular measure than Pass@1.

Agent Profile	Pass@1 (Core)	Pass@1 (Robust.)	$\mathcal{R}_R$ Score $\uparrow$
Agent X (Robust)	62.5%	41.3% (-21.2)	0.81
Agent Y (Brittle)	58.8%	38.1% (-20.7)	0.65

Table 6: Ablation study of the TRACE-Scaffolding set. λ_{min} provides a clear ranking where Pass@1 fails.

Agent	Pass@1 $\uparrow$	Avg. MHR $\bar{\lambda}_{min}$ $\downarrow$
AgentFounder-30B	4.0%	0.22
WebSailor-V2-30B	2.0%	0.28
DeepSeek-V3.1-671B	4.0%	0.35

Table 7: Diagnostic analysis of agent “personalities” on TRACE-Core. The correlation between Trajectory Reproducibility Score (TRS) and Entropy Adaptability ($\mathcal{E}_A$) reveals distinct strategic profiles that answer RQ3.

Agent Profile	TRS $\uparrow$	$\mathcal{E}_A$ $\uparrow$	Inferred Profile
<i>SOTA Agents</i>			
AgentFounder-30B	0.89	0.82	Systematic & Efficient
WebSailor-V2-30B	0.78	0.88	Adaptive & Rational
DeepSeek-V3.1-671B	0.83	0.68	Consistent but Inefficient
<i>Baseline Agent</i>			
ReAct (Qwen-30B)	0.62	0.55	Heuristic & Unstable

5 CONCLUSION & OUTLOOK

In this paper, we introduced TRACE (Trajectory-Aware Comprehensive Evaluation), a novel evaluation framework designed to address the critical limitations of traditional, outcome-based metrics for Deep Research Agents. Our framework moves beyond singular accuracy scores, using a Hierarchical Trajectory Utility Function to holistically evaluate process quality—including efficiency and robustness—and diagnostic tools like Scaffolded Capability Assessment to measure latent attributes. Our experiments demonstrate that TRACE successfully unveils the “high-score illusion”, revealing that top-performing agents are not always the most efficient or reliable. By providing a rigorous, multi-dimensional analysis, TRACE offers a principled path for developing and comparing truly reliable Deep Research Agents.

Looking ahead, its utility function can serve as a reward signal for agent optimization, and we plan to extend the framework to other complex, long-horizon domains.

Acknowledgments

The research presented in this paper was partially supported by the Research Grants Council of the Hong Kong Special Administrative Region, China (CUHK 2410072, RGC R1015-23) and (CUHK 2300246, RGC C1043-24G).

References

- [1] Heeseung Bang and Andreas A. Malikopoulos. 2023. Optimal trajectory planning meets network-level routing: Integrated control framework for emerging mobility systems. *arXiv:2311.13193* [cs.LG]
- [2] Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mohammad Bavarian, Clemens Winter, Philippe Tillet, Filipe de Avila Belbute Peres, Fps, L. W. He, D. Li, Y. Tian, Z. Wang, and J. Li. 2021. Evaluating Large Language Models Trained on Code. *arXiv:2107.03484* [cs.LG]
- [3] Clayton Cohn, Surya Rayala, Namrata Srivastava, Joyce Horn Fonteles, Shruti Jain, Xinying Luo, Divya Mereddy, Naveeduddin Mohammed, and Gautam Biswas. 2025. A Theory of Adaptive Scaffolding for LLM-Based Pedagogical Agents. *arXiv:2508.01503* [cs.CL]
- [4] Mingxuan Du1, 2 Benfeng Xu1, Chiwei Zhu1, Xiaorui Wang2, and Zhendong Mao. 2025. DeepResearch Bench: A Comprehensive Benchmark for Deep Research Agents. *arXiv:2506.11763* [cs.CL]
- [5] Jiazhan Feng, Shijue Huang, Xingwei Qu, Ge Zhang, Yujia Qin, Baoquan Zhong, Chengquan Jiang, Jinxin Chi, and Wanjuan Zhong. 2025. ReTool: Reinforcement Learning for Strategic Tool Use in LLMs. *arXiv:2504.11536* [cs.CL]
- [6] Renju Feng, Ning Xi, Duanfeng Chu, Rukang Wang, Zejian Deng, Anzheng Wang, and Liping Lu. 2025. ARTEMIS: Autoregressive End-to-End Trajectory Planning with Mixture of Experts for Autonomous Driving. *arXiv:2504.19580* [cs.LG]
- [7] Yunfan Gao, Yun Xiong, Yijie Zhong, Yuxi Bi, Ming Xue, and Haofen Wang. 2025. Synergizing RAG and Reasoning: A Systematic Review. *arXiv:2504.15909* [cs.IR]
- [8] Xinyu Geng, Peng Xia, Zhen Zhang, and Xinyu Wang. 2025. Web-Watcher: Breaking New Frontier of Vision-Language Deep Research Agent. *arXiv:2508.05748* [cs.CL]
- [9] Zhihan Guo, Feiyang Xu, Yifan Li, Muzhi Li, Shuai Zou, Jiele Wu, Han Shi, Haoli Bai, Ho-fung Leung, and Irwin King. 2025. Dr. Mi-Bench: A Modular-integrated Benchmark for Scientific Deep Research Agent. *arXiv preprint arXiv:2512.00986* (2025).
- [10] Yuxuan Huang†1, Yihang Chen†2, Haozheng Zhang†2, Kang Li3, Huichi Zhou4, Meng Fang1, Linyi Yang4, Xiaoguang Li2, Lifeng Shang2, Songcen Xu2, Jianye Hao2, Kun Shao‡2, and Jun Wang‡. 2025. DEEP RESEARCH AGENTS: A SYSTEMATIC EXAMINATION AND ROADMAP. *arXiv:2506.18096* [cs.CL]
- [11] Abhinav Java, Ashmit Khandelwal, Sukruta Midigeshi, Aaron Halfaker, Amit Deshpande, Navin Goyal, Ankur Gupta, Nagarajan Natarajan, and Amit Sharma. 2025. Characterizing deep research: A benchmark and formal definition. *arXiv:2508.04183* [cs.CL]
- [12] Jiyue Jiang, Pengan Chen, Liheng Chen, Sheng Wang, Qinghang Bao, Lingpeng Kong, Yu Li, and Chuan Wu. 2025. How Well Do LLMs Handle Cantonese? Benchmarking Cantonese Capabilities of Large Language Models. In *Findings of the Association for Computational Linguistics: NAACL 2025*, Luis Chiruzzo, Alan Ritter, and Lu Wang (Eds.). Association for Computational Linguistics, Albuquerque, New Mexico, 4464–4505. doi:10.18653/v1/2025.findings-naacl.253
- [13] Jiyue Jiang, Pengan Chen, Jiuming Wang, Dongchen He, Ziqin Wei, Liang Hong, Licheng Zong, Sheng Wang, Qinzhe Yu, Zixian Ma, et al. 2025. Benchmarking large language models on multiple tasks in bioinformatics nlp with prompting. *arXiv preprint arXiv:2503.04013* (2025).
- [14] Jiyue Jiang, Yunke Li, Shiwei Cao, Yuheng Shan, Yuexing Liu, Tianyi Fei, Yule Yu, Yi Feng, Yu Li, Yixue Li, and Jiao Yuan. 2025. Artificial intelligence in bioinformatics: a survey. *Briefings in Bioinformatics* 26, 6 (11 2025), bbaf576. *arXiv:https://academic.oup.com/bib/article-pdf/26/6/bbaf576/65235101/bbaf576.pdf* doi:10.1093/bib/bbaf576
- [15] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *arXiv:2005.11401* [cs.CL]
- [16] Kuan Li, Zhongwang Zhang, Huifeng Yin, Rui Ye, Yida Zhao, Liwen Zhang, Litu Ou, Dingchu Zhang, Xixi Wu, Jialong Wu, Xinyu Wang, Zile Qiao, Zhen Zhang, Yong Jiang, Pengjun Xie, Fei Huang, and Jingren Zhou. 2024. WebSailor-V2: Bridging the Chasm to Proprietary Agents via Synthetic Data and Scalable Reinforcement Learning. *arXiv:2409.13305* [cs.LG]
- [17] Kuan Li, Yida Zhao, Huifeng Yin, Liwen Zhang, Zhongwang Zhang, Litu Ou, Rui Ye, Yifan Song, Xinyu Wang, Jialong Wu, Zile Qiao, Zhen Zhang, Xixi Wu, Yong Jiang, Pengjun Xie, Fei Huang, Teng Su, and Jingren Zhou. 2024. WebSailor: Navigating Super-human Reasoning for Web Agent. *arXiv:2407.02592* [cs.CL]
- [18] Xiaoxi Li1, Jijie Jin1, Guanting Dong1, Hongjin Qian2, Yutao Zhu1, Yongkang Wu3, Ji-Rong Wen1, and Zhicheng Dou1. 2025. Webthinker: Empowering large reasoning models with deep research capability. *arXiv:2504.21776* [cs.CL]
- [19] Jiahong Liu, Zexuan Qiu, Zhongyang Li, Quanyu Dai, Wenhao Yu, Jieming Zhu, Minda Hu, Menglin Yang, Tat-Seng Chua, and Irwin King. 2025. A survey of personalized large language models: Progress and future directions. *arXiv preprint arXiv:2502.11528* (2025).
- [20] Zeyuan Liu, Sitong Cheng, Yushi Sun, Zecheng Wang, Haotian Liu, Yangyi Chen, Wenbiao Ding, Zhongjin Zhai, Yifei Shen, Chaowei Xiao, Chunyuan Li, and Yuxiao Dong. 2024. WebSUPER: A Benchmark for Sequential Program-Enhanced Reasoning in Web Agents. *arXiv:2406.11581* [cs.AI]
- [21] Ziru Liu, Cheng Gong, Xinyu Fu, Yaofang Liu, Ran Chen, Shoubo Hu, Suiyuan Zhang, Rui Liu, Qingfu Zhang, and Dandan Tu. 2024. GHPO: Adaptive Guidance for Stable and Efficient LLM Reinforcement Learning. *arXiv:2407.10628* [cs.CL]
- [22] Utsav Maskey, Sumit Yadav, Mark Dras, and Usman Naseem. 2025. SafeConstellations: Steering LLM Safety to Reduce Over-Refusals Through Task-Specific Trajectory. *arXiv:2508.11290* [cs.CL]
- [23] Grégoire Mialon, Clémentine Fourrier, Thomas Wolf, Craig Swift, Yann LeCun, and Thomas Scialom. 2023. GAIA: a benchmark for General AI Assistants. *arXiv:2311.12983* [cs.CL]
- [24] Jonathan Michaux, Seth Isaacson, Challen Ennifful Adu, Adam Li, Rahul Kashyap Swayampakula, Parker Ewen, Sean Rice, Katherine A. Skinner, and Ram Vasudevan. 2024. Let's Make a Splan: Risk-Aware Trajectory Optimization in a Normalized Gaussian Splat. *arXiv:2409.16915* [cs.RO]
- [25] Agada Joseph Oche, Ademola Glory Folashade, Tirthankar Ghosal, and Arpan Biswas. 2025. A Systematic Review of Key Retrieval-Augmented Generation (RAG) Systems: Progress, Gaps, and Future Directions. *arXiv:2507.18910* [cs.IR]
- [26] Vardaan Pahuja, Yadong Lu, Corby Rosset, Boyu Gou, Arindam Mitra, Spencer Whitehead, Yu Su, and Ahmed Awadallah. 2025. Explorer: Scaling Exploration-driven Web Trajectory Synthesis for Multimodal Web Agents. *arXiv:2502.11357* [cs.AI]
- [27] Cheng Qian, Emre Can Acikgoz, Qi He, Hongru Wang, Xiushi Chen, Dilek Hakkani-Tür, Gokhan Tur, and Heng Ji. 2025. ToolRL: Reward is All Tool Learning Needs. *arXiv:2507.13958* [cs.CL]
- [28] Junlong Ren, Gangjian Zhang, Honghao Fu, Pengcheng Wu, and Hao Wang. 2025. WaMo: Wavelet-Enhanced Multi-Frequency Trajectory Analysis for Fine-Grained Text-Motion Retrieval. *arXiv:2508.03343* [cs.CV]
- [29] Aditi Singh, Abul Ehtesham, Saket Kumar, and Tala Talaei Khoei. 2025. AGENTIC RETRIEVAL-AUGMENTED GENERATION: A SURVEY ON AGENTIC RAG. *arXiv:2507.18910* [cs.AI]
- [30] Yifan Song, Kuan Li, Zile Qiao, Xinyu Wang, Rui Ye, Jialong Wu, Yong Jiang, Pengjun Xie, Fei Huang, and Jingren Zhou. 2024. ReSum: A Resource-Efficient Summarization Framework for Long-Context LLMs. *arXiv:2407.02597* [cs.CL]
- [31] Zhaochen Su, Linjie Li, Mingyang Song, Yunzhuo Hao, Zhengyuan Yang, Jun Zhang, Guanjie Chen, Jiawei Gu, Juntao Li, Xiaoye Qu, and Yu Cheng. 2025. OPENTHINKING: Learning to Think with Images via Visual Tool Reinforcement Learning. *arXiv:2505.08617* [cs.CL]
- [32] Zhong-Shen Sun, Jia-Qi Li, Hong-Bin YU, Ting-En Lin, and Chen-Bin YU. 2024. WebShaper: A Self-Correction Agent for Web Navigation. *arXiv:2407.01633* [cs.AI]
- [33] Guiyao Tie, Xueyang Zhou, Tianhe Gu, Ruihang Zhang, Chaoran Hu, Sizhe Zhang, Mengqu Sun, Yan Zhang, Pan Zhou, and Lichao Sun. 2025. MMLU-Reason: Benchmarking Multi-Task Multi-modal Language Understanding and Reasoning. *arXiv:2505.16459* [cs.CL]
- [34] Ada Defne Tur, Nicholas Meade, Dian Yu, Xing Han Lu, Alejandra Zambrano, Arkil Patel, and Esin Durmus. 2025. SAFEARENA: Evaluating the Safety of Autonomous Web Agents. *arXiv:2503.04957* [cs.LG]
- [35] Zhendong Wang, Jianxiang Li, Haoyi Niu, Jingjing Liu, Weidong He, and Ding Li. 2023. DAPO: Diffusion-Based Offline Reinforcement Learning. *arXiv:2306.17489* [cs.LG]
- [36] Jason Wei, Maarten Bosma, Vincent Y. Zhao, Kelvin Guu, Adams Wei Yu, Brian Lester, Nan Du, Andrew M. Dai, and Quoc V. Le. 2019. Finetuned Language Models Are Zero-Shot Learners. *arXiv:19.01652* [cs.CL]
- [37] Jialong Wu, Wenbiao Yin, Yong Jiang, Zhenglin Wang, Zekun Xi, Runnan Fang, Linhai Zhang, Yulan He, Deyu Zhou, Pengjun Xie, and Fei Huang. 2025. WebWalker: Benchmarking LLMs in Web Traversal. *arXiv:2501.07572* [cs.CL]
- [38] Yifan Wu, Jiyue Jiang, Xichen Ye, Yiqi Wang, Chang Zhou, Yitao Xu, Jiayang Chen, He Hu, Weizhong Zhang, Cheng Jin, et al. 2025. Investigating Data Pruning for Pretraining Biological Foundation Models at Scale. *arXiv preprint arXiv:2512.12932* (2025).
- [39] RENJUN XU and JINGWEN PENG. 2025. A Comprehensive Survey of Deep Research: Systems, Methodologies, and Applications. *arXiv:2506.12594* [cs.AI]
- [40] Lujie Yang, H.J. Terry Suh, Tong Zhao, Bernhard Paus, Græsdal, Tarik Keleştemur, Jiuguang Wang, Tao Pang, , and Russ Tedrake. 2025. Physics-Driven Data Generation for Contact-Rich Manipulation via Trajectory Optimization. *arXiv:24502.20382* [cs.RO]
- [41] Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. 2018. HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Brussels, Belgium, 2369–2380. doi:10.18653/v1/D18-1259
- [42] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2022. ReAct: Synergizing Reasoning and Acting in Language Models. *arXiv:2210.03629* [cs.CL]

- [43] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2022. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629 [cs.CL]
- [44] Huilin Yin, Yangwenhui Xu, Jiaxiang Li, Hao Zhang, and Gerhard Rigoll. 2025. Knowledge-Informed Multi-Agent Trajectory Prediction at Signalized Intersections for Infrastructure-to-Everything. arXiv:2501.13461 [cs.RO]
- [45] Dianzhi Yu, Conghao Xiong, Yankai Chen, Wenqian Cui, Xinni Zhang, Yifei Zhang, Hao Chen, Joseph J. Y. Sung, and Irwin King. 2026. ConSurv: Multimodal Continual Learning for Survival Analysis. arXiv:2511.09853 [cs.LG] <https://arxiv.org/abs/2511.09853>
- [46] Dianzhi Yu, Xinni Zhang, Yankai Chen, Aiwei Liu, Yifei Zhang, Philip S. Yu, and Irwin King. 2024. Recent Advances of Multimodal Continual Learning: A Comprehensive Survey. arXiv:2410.05352 [cs.LG] <https://arxiv.org/abs/2410.05352>
- [47] Yao-Ching Yu, Hung-Ting Chen, Chia-Hsuan Lee, and Yun-Nung Chen. 2024. WebWalker: A Disentangled and Reasoning-Enhanced Agent for Web Navigation. arXiv:2405.15848 [cs.CL]
- [48] Zhongwang Zhang, Kuan Li, Yida Zhao, Huifeng Yin, Zhen Zhang, Litu Ou, Liwen Zhang, Rui Ye, Yong Jiang, Pengjun Xie, Fei Huang, and Jingren Zhou. 2024. Scaling Agents via Continual Pre-training. arXiv:2409.13310 [cs.CL]
- [49] Yuxiang Zheng, Dayuan Fu, Xiangkun Hu, Xiaojie Cai, Lyumanshan Ye1, Pengrui Lu1, and Pengfei Liu. 2025. Browsecomp-plus: A more fair and transparent evaluation benchmark of deep-research agent. arXiv:2504.03160 [cs.CL]
- [50] Yuxiang Zheng, Dayuan Fu, Xiangkun Hu, Xiaojie Cai, Lyumanshan Ye1, Pengrui Lu1, and Pengfei Liu. 2025. Deepresearcher: Scaling deep research via reinforcement learning in real-world environments. arXiv:2504.03160 [cs.CL]

A Appendix

A.1 RELATED WORKS

Our work is situated at the intersection of two rapidly evolving research areas: the evaluation of Deep Research Agents and the trajectory-level analysis of sequential decision-making processes.

A.1.1 Evaluation of Deep Research Agents. The evaluation of Deep Research Agents has advanced through challenging benchmarks like **BrowseComp-en/zh** [16, 30, 48] and **GAIA** [17, 20, 47], which test the limits of SOTA agents such as AgentFounder [48], WebSailor-V2 [16], and WebShaper [20]. However, this paradigm is fundamentally limited by its reliance on singular, outcome-based metrics like Pass@1, a practice prevalent across much of the recent agent literature [17, 30, 32, 48]. This creates a “**high-score illusion**” [33], where the quality, efficiency, and robustness of the reasoning process are ignored. Such an approach makes it impossible to distinguish between an agent that intelligently solves a problem and one that arrives at the correct answer through a lucky or inefficient path, thereby masking crucial deficiencies in planning and trustworthiness [17, 30]. **TRACE addresses this critical gap by shifting the evaluation from final outcomes to a holistic, trajectory-aware perspective.**

A.1.2 Trajectory-level Analysis. Trajectory-level analysis is central to modern agent *training*, particularly in Reinforcement Learning (RL), where algorithms like DAPO [35] and GHPO [21] are fundamentally process-aware, optimizing the entire decision-making sequence. This focus on process is a key element in the post-training pipelines of many SOTA agents that use RL to refine their behaviors [16, 48]. Despite its importance in training, this process-aware perspective is largely absent in agent *evaluation*, which typically treats agents as black boxes by focusing only on final scores [17, 20, 47]. This leaves researchers unable to diagnose *why* an agent succeeds or fails [4]—whether the cause is poor planning, flawed reasoning, or an inability to recover from errors. **Our work**

bridges this gap by being the first to systematically adapt principles from RL into a formal evaluation suite. TRACE introduces novel diagnostic tools[5], such as Scaffolded Capability Assessment [3] and Entropy Adaptability [35], to move beyond measuring *what* an agent can do, and begin to understand *how* and *why* it performs, drawing conceptual inspiration from the dynamics explored in advanced RL frameworks [21, 35].

A.2 Rigorous Mathematical Justification of TRACE Metrics

A.2.1 Justification for the Hierarchical Trajectory Utility Function.

Theorem 1. The weighted geometric mean (GM) exhibits unbounded sensitivity to a component score approaching zero, unlike the constant sensitivity of the weighted arithmetic mean (AM).

Proof. Let component scores be $\mathbf{x} = \{x_1, \dots, x_n\}$, weights $\omega_i > 0$, $\sum \omega_i = 1$. The AM is $A(\mathbf{x}) = \sum_i \omega_i x_i$. The GM is $G(\mathbf{x}) = \prod_i x_i^{\omega_i}$. The sensitivity of AM with respect to a component x_k is its partial derivative:

$$\frac{\partial A}{\partial x_k} = \omega_k \quad (\text{Constant Sensitivity})$$

The sensitivity of GM is:

$$\frac{\partial G}{\partial x_k} = \omega_k x_k^{\omega_k - 1} \prod_{i \neq k} x_i^{\omega_i} = \omega_k \frac{G(\mathbf{x})}{x_k}$$

The ratio of sensitivities is $\frac{\partial G / \partial x_k}{\partial A / \partial x_k} = \frac{G(\mathbf{x})}{x_k}$. As a component fails, $x_k \rightarrow 0^+$, this ratio approaches infinity:

$$\lim_{x_k \rightarrow 0^+} \frac{G(\mathbf{x})}{x_k} = \infty$$

This proves the GM’s sensitivity is infinitely greater than the AM’s near critical failure points, validating its use for the “weakest link” principle. ■

A.2.2 Justification for Process Efficiency $\mathcal{E}(\mathcal{H})$.

Proposition 2.1. The logarithmic cost functional $J(\mathcal{H})$ exhibits diminishing sensitivity, correctly modeling the decreasing marginal impact of costs.

Proof. Let total raw cost be $C_{tot} = \sum_{t=1}^T C(a_t) p_t$. The cost functional is $J(C_{tot}) = 1 + \ln(C_{tot})$. The sensitivity is the first derivative:

$$\frac{dJ}{dC_{tot}} = \frac{1}{C_{tot}}$$

The rate of change of sensitivity is the second derivative:

$$\frac{d^2 J}{dC_{tot}^2} = -\frac{1}{C_{tot}^2} < 0$$

Since the second derivative is always negative for $C_{tot} > 0$, the function is strictly concave. This formally proves that the functional exhibits diminishing sensitivity: each additional unit of cost contributes less to the total functional value than the one before it. ■

Proposition 2.2. The Marginal Information Gain g_t is positive if and only if the relevance of the current observation is strictly greater than the supremum of all prior relevances.

Proof. Let relevance $r_t = \cos(\Phi(o_t), \Phi(A_{g_t}))$ and historical maximum $S_{t-1} = \sup_{i < t} \{r_i\}$. The MIG is $g_t = \max(0, r_t - S_{t-1})$. ($\Rightarrow$) Assume $g_t > 0$. By definition of max, it must be that $r_t - S_{t-1} > 0$, which implies $r_t > S_{t-1}$. ($\Leftarrow$) Assume $r_t > S_{t-1}$. Then the difference $r_t - S_{t-1}$ is positive. Therefore, $g_t = r_t - S_{t-1} > 0$. The biconditional holds. ■

A.2.3 Justification for Cognitive Quality $C(\mathcal{H})$.

Proposition 3.1. The Evidence Grounding score $\mathcal{G}_E$ has unbounded sensitivity to any single entailment probability p_k approaching zero.

Proof. The score is $\mathcal{G}_E(\mathbf{p}) = (\prod_{i=1}^N p_i)^{1/N}$. We use logarithmic differentiation.

$$\ln(\mathcal{G}_E) = \frac{1}{N} \sum_{i=1}^N \ln(p_i)$$

Differentiating with respect to p_k :

$$\frac{1}{\mathcal{G}_E} \frac{\partial \mathcal{G}_E}{\partial p_k} = \frac{1}{N} \frac{1}{p_k}$$

The sensitivity is thus:

$$\frac{\partial \mathcal{G}_E}{\partial p_k} = \frac{\mathcal{G}_E}{N p_k}$$

As any claim becomes ungrounded, $p_k \rightarrow 0^+$, and the sensitivity $\frac{\partial \mathcal{G}_E}{\partial p_k} \rightarrow \infty$. This proves extreme sensitivity to hallucination. ■

Proposition 3.2. The Reasoning Robustness score $\mathcal{R}_R$ is a normalized, strictly monotonically decreasing function of the average recovery latency $\bar{D}$.

Proof. The score is $\mathcal{R}_R(\bar{D}) = \exp(-\lambda \bar{D})$, with $\lambda > 0$ and $\bar{D} \geq 0$.
1. Normalization (Bounds): The function’s value at zero latency is $\mathcal{R}_R(0) = \exp(0) = 1$. The limit as latency grows is $\lim_{\bar{D} \rightarrow \infty} \exp(-\lambda \bar{D}) = 0$. The range is thus $\mathcal{R}_R \in (0, 1]$, which is a valid normalized score range.
2. Monotonicity: The derivative with respect to $\bar{D}$ is:

$$\frac{d\mathcal{R}_R}{d\bar{D}} = -\lambda \exp(-\lambda \bar{D})$$

Since $\lambda > 0$ and $\exp(\cdot)$ is always positive, the derivative $\frac{d\mathcal{R}_R}{d\bar{D}} < 0$ for all $\bar{D} \geq 0$. Therefore, $\mathcal{R}_R$ is a strictly decreasing function of average latency. ■

A.3 Implementation Details

Our entire experimental pipeline is designed for rigor and reproducibility. All agent evaluations were conducted within a unified execution framework to enforce a level playing field and ensure a fair comparison across different models. This framework, built upon the open-source ‘Qwen-Agent’ library [47], standardizes the entire agent-environment interaction loop, including the action space (e.g., ‘Search’, ‘Finish’) and the format of observations returned to the agent. Open-source models were run from their official codebases within this environment, while proprietary systems were accessed via their APIs and integrated through a standardized wrapper. The experiments were conducted on a high-performance cluster of 8x NVIDIA A100 (80GB) GPUs, a setup consistent with large-scale agent training experiments [21]. For all agent inferences, we used a consistent set of parameters: a temperature of 0.85 and a top-p of 0.95, which are in line with hyperparameter settings used in recent SOTA agent evaluations [16, 30]. To prevent infinite loops, we set a maximum of 60 tool calls per task [30]. The calculation of our TRACE metrics was also strictly configured: final answer correctness for $\mathcal{A}(\mathcal{H})$ is determined using an LLM-as-Judge approach with GPT-4-Turbo, a standard practice in recent agent benchmarks [16]; all semantic similarity calculations use the ‘all-mpnet-base-v2’ encoder; and entailment probabilities for Evidence Grounding ($\mathcal{G}_E$) were computed using ‘DeBERTa-v3-large’. The hyperparameters for our metric formulas were fixed as follows: $\omega_E = 0.5$, $\omega_C = 0.5$, $\alpha = 0.5$, $\gamma = 0.05$, $\beta = 0.5$, and $\lambda = 0.1$. For our diagnostic tools, θ_{succ} was set to 0.9 (over 10 runs), and the TRS was calculated using $K = 5$ runs per task [2, 16].